

Ahmed Mahmoud Abbas

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🛡 Military Status: Exempted 🔗 LinkedIn 🐙 GitHub 📁 Portfolio

Profile

Backend and DevOps Engineer with hands-on production experience in banking infrastructure and security tooling. Built ACR-QA, a SAST platform outperforming Snyk and Semgrep on RealVuln 2026 with exploit-verified, ECDSA-attested findings at \$0 recurring cost. Delivered 4 production-handoff prototypes at NBK Egypt across AI/ML, banking infrastructure, and secure document delivery. Engineered a Kubernetes observability stack at 225 req/sec, 62ms P95 under Egypt's national DEPI program. Available for remote and on-site backend/DevOps roles from July 2026.

Professional Experience

DevOps Engineering Trainee 06/2025 – 12/2025
Digital Egypt Pioneers Initiative (MCIT – EYouth). Zamalek, Giza, Egypt

- Selected for a national DevOps training program sponsored by **MCIT**.
- Automated CI/CD pipelines with Git, Jenkins, and Docker across a **5-stage GitHub Actions workflow**
- Built reusable IaC modules with **Terraform** and **Ansible** to provision **5+ environments**.
- Deployed, monitored, and optimized containerized microservices on **Kubernetes** clusters using **Prometheus** and **Grafana** for observability
- Capstone: production URL shortener with **225 req/sec throughput, 99.93% uptime, P95 latency 62ms** – 19 Grafana panels across 3 dashboards, Kubernetes HPA, Terraform IaC, 5-stage CI/CD.

R&D Intern – Development & Application 08/2025 – 09/2025
National Bank of Kuwait - Egypt New Cairo, Cairo, Egypt

- Built and delivered AI/ML prototypes for NBK Egypt's R&D division – including a **loan risk predictor** (Streamlit, RandomForest, SMOTE) and an **OFAC sanctions screening system** (Python, TF-IDF, fuzzy logic).
- Developed a **secure PDF retrieval service** (Spring Boot, RSA/AES) integrated with **Temenos T24** via Tomcat.
- Built **Outlook→SFTP automation** (Java, JACOB COM bridge, JSch) to extract and securely transfer email attachments to Linux servers, replacing a manual banking workflow.
- Applied SOLID principles, design patterns, and Docker containerization across all **4 prototypes**.
- Internship extended by **2 weeks based on performance** – all 4 prototypes delivered as **production handoff artifacts** with RUNBOOK documentation.

Information Technology Intern 08/2024 | Sheikh Zayed, Giza, Egypt
Emkay Foods

- Supported IT operations across **150+ endpoints**: Active Directory, DHCP, VPN/VLAN configuration, and managed switch/firewall setup (**FortiGate, Sophos**)
- Assisted with ERP user provisioning and access management, improving onboarding and ticket turnaround

Projects

ACR-QA – Automated Code Review Platform 🔗 12/2025 – 05/2026

Graduation Project · Python · RAG · Docker · PostgreSQL · GitHub Actions

- Built a language-agnostic SAST platform running **19 detection engines** in parallel across Python, JS/TS, and Go – canonical normalization schema across all tools, **327+ rule mappings**, 9/10 OWASP Top 10 covered
- Achieved **96.4% Confirmed Tier precision** and **100% CVE recall** (8/8); **F1 98.2%** vs Bandit 21.8% / Semgrep 45.7%; beats Snyk and Semgrep CE on **RealVuln 2026** full-corpus recall
- Exploit-verified findings** via ephemeral Docker sandbox (13 exploit categories – SQLi, CMDi, SSTI); every verdict **ECDSA-P256 signed**, logged to Sigstore Rekor, **SLSA L3 provenance** and post-quantum Dilithium3 attestation
- RAG-grounded** AI explanations from a 66-rule knowledge base with **semantic entropy scoring** (3× LLM runs) and hallucination detection – \$0 recurring cost via Groq
- 3,247 tests at 88% core coverage**; GitHub Actions + GitLab CI; SARIF v2.1.0 export; auto-block merge gate via **policy engine**

Redis from Scratch (Rust, TCP, Multithreading, Concurrency, Systems Programming)

03/2025 – 06/2025

- Built a **multithreaded in-memory key-value store** with support for Strings, Lists, Sets, Sorted Sets, and TTL.
- Implemented the command parsing layer supporting **16 Redis commands** across **4 data structures** including TTL-based key expiry and LRANGE/ZRANGE with index bounds.
- Ensured concurrency safety and atomic operations using **Arc, Mutex**, and Rust's ownership model.

Containerized URL Shortener – Production Monitoring

10/2025 – 11/2025

Stack

DEPI Graduation Project · Node.js · Docker · Kubernetes ·

Prometheus · Grafana · GitHub Actions · Terraform

- Built and containerized a URL shortener REST API (Node.js + Express + SQLite) with custom Prometheus metrics instrumentation – **19 Grafana panels** across **3 dashboards** (main monitoring, analytics, system health)
- Achieved **225 req/sec throughput**, **99.93% uptime**, and **P95 latency of 62ms** under k6 load testing with **75 concurrent users**
- Implemented full DevOps lifecycle: **5-stage GitHub Actions CI/CD**, Kubernetes deployment with **HPA** and PodDisruptionBudgets, **Terraform IaC** modules, and multi-channel alerting (Slack, Email, Discord)
- Added automated disaster recovery with **15-minute RTO**, Docker volumes for persistence, and a security audit (**B+ grade**, zero critical vulnerabilities)

DecodedSound – AI Cultural Translation Engine

02/2026 – 05/2026

Next.js 15 · TypeScript · Groq LLaMA · Whisper · PostgreSQL ·

Prisma · Docker

- Engineered a **3-input pipeline** (typed lyrics, audio upload, YouTube URL via yt-dlp) → Groq Whisper transcription → **LLaMA 3.3 70B** translation → **4-panel output** (plain translation, line-by-line annotation, slang dictionary, cultural context).
- Built a full-stack AI translation platform for SDK (Esdeekid) music – a hyper-coded Cape Flats dialect mixing Kaaps Afrikaans, Xhosa, and prison gang codes that no existing AI model handles correctly.
- Built a living knowledge base of **502 hand-curated slang terms** with community contributions, AI candidate discovery, admin moderation panel, and auto-downgrade on **3 community flags**.
- Implemented **12 REST API routes**, IP-based **sliding-window rate limiting** across 3 tiers, YouTube caching (translated URLs served instantly on repeat).

Education

BSc in Computer Science – Software Engineering

10/2022 – 06/2026 | South Sinai, Egypt

King Salman International University

- Activities: IEEE, GDSC, ICPC communities, COP27 Youth Volunteer, Huawei ICT Academy
- Achievement: ICPC ECPC 2024 – Honorable Mention, 2nd place among KSIU teams

Skills

- **Languages:** Rust, C++, Python, Java, TypeScript, JavaScript
- **Backend:** Node.js, Express, FastAPI, Flask, Spring Boot, REST APIs, PostgreSQL, Redis, MongoDB, SQLite, Prisma, JWT, OAuth
- **DevOps:** Docker, Docker Compose, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, GitLab CI, Prometheus, Grafana, Nginx, CI/CD pipelines
- **ML & AI:** scikit-learn, RAG pipelines, Ollama, Groq, LLaMA, Whisper, SMOTE, PCA, RandomForest, Pandas, NumPy, Streamlit
- **Security & AppSec:** SAST, exploit verification, ECDSA/cryptographic attestation, SLSA supply chain security, Sigstore/Cosign, RSA/AES encryption, SFTP, CWE mapping, OWASP Top 10, Semgrep, Bandit, Ruff
- **Infra & Networking:** Linux, AWS, GCP, SSH, systemd, TCP/IP, DNS, DHCP, VLANs, VPNs, NAT, FortiGate, Sophos, Active Directory
- **Testing & Observability:** pytest, k6, Apache Bench, SARIF v2.1.0, Grafana dashboards, load testing, Docker sandboxing
- **Tools & Practices:** Git, OOP, data structures, concurrency, multithreading, microservices, SOLID principles, clean architecture, CI/CD, RAG pipeline design

Languages

Arabic
Mother Tongue

English
Fluent